

WORK EXPERIENCE

- *Data Scientist at 9D Technologies (Feb 2026–Now):*

- Architected and implemented the **multi-agent orchestration layer** for a Google ADK-based search system, enabling query decomposition, parallel specialist-agent execution, result aggregation, and final-answer synthesis across Google, Tavily, and Brave Search.
- Took the multi-agent search system from internal prototype to an **AWS Marketplace-ready deployment**, handling containerization, dependencies, configuration, tool integrations, edge cases, and packaging; also packaged a separate real-time ASR pipeline with speaker diarization and identification for Marketplace deployment.
- Built a systematic evaluation pipeline for **50+ quantized language models (GGUF)** for on-device Android deployment, combining manual testing, DeepEval with custom datasets and metrics, and 20+ public benchmarks across six domains; recommended ~25 models (4–5 per domain) ultimately used in the app across device tiers.
- Optimized **system prompts, sampling parameters, and on-device safety guardrails** for domain-specific assistants, developing and training a TensorFlow safety/disclosure classifier and deploying it as a TFLite model across education, coding, research, mathematics, healthcare, and finance use cases.
- Developed a **domain-specific speech-to-speech evaluation framework** for a female-health AI assistant, extending Eva-Bench with custom scenarios and evaluation metrics for medical safety, red-flag escalation, unsupported claims, conciseness, opening/closing quality, relevant follow-ups, and turn-taking latency; benchmarked PersonaPlex against GPT Realtime models and informed the selection of GPT Realtime 2.1 Mini for initial deployment.
- Conducted **GPU-hosting vs API cost analysis** for the speech-to-speech system, balancing model quality, latency, expected demand, and infrastructure costs to inform deployment strategy and potential future migration to self-hosted inference.

- *Data Scientist at Afiniti (Sep 2024–Feb 2026):*

- Optimized real-time **agent-customer pairing models** serving 200,000+ customers per week, using Bayesian agent segmentation models in Stan and LightGBM/XGBoost caller segmentation models to improve call-center matching.
- Consistently achieved **1.5x–2x target gains** in client-defined pairing performance, increasing sales and customer retention relative to clients' existing agent-customer pairing methods.
- Led **migration from R to Python for production pipelines**, enabling more extensive hyperparameter tuning, validation, visualization, and adoption of advanced ML approaches while maintaining existing production functionality.
- Built **real-time production monitoring systems** with SQL-based metrics, dashboards, and alerts to track pairing quality, agent utilization, caller wait times, and other operational indicators and proactively identify model or deployment issues.
- Led **ONNX pipeline development** for caller segmentation models, adapting model evaluation and production workflows to support ONNX-based inference for improved scalability.
- Collaborated with **cross-functional teams** to translate statistical and machine learning models into production call-center optimization systems and address real-world operational constraints.

- *Teaching Assistant at Lahore University of Management Sciences (Jan 2024–Jun 2024):* Co-instructed a **graduate-level course on Convex Optimization** with Dr. Hassan Mohy Ud Din.

RESEARCH EXPERIENCE

- *Under Dr. Kim-Chuan Toh (Oct 2025–Now):*

- Reimplemented the **ADMM component of the SDPNAL+ solver** from MATLAB to Julia for cross-language reproducibility and extensibility.
- Ported the solver from CPU to GPU using CUDA with **custom kernels and restructured data layouts** for efficient GPU utilization.
- Benchmarked implementations across 30+ test problems, **demonstrating 2-30x speedups** on GPU over CPU for medium-to-large instances.
- Restructured MATLAB-specific constructs (e.g., cell arrays, MEX dependencies) into Julia-native designs, improving **computational efficiency and maintainability**.
- Evaluated an alternative **PSD projection method** to replace the costly eigendecomposition-based approach.
- Diagnosed and resolved numerical instability in failed factorizations, identifying **ill-conditioning from extreme spectral scaling** as the root cause.

- *Under Dr. Hassan Mohy Ud Din (Jun 2024-Jun 2026):*
 - Conducted **end-to-end research**, including problem formulation, literature review, experimental design, and scientific writing ([Wave-GMS](#)).
 - Designed and implemented **computer vision models for medical image segmentation** using diffusion-based and variational approaches.
 - Built **training and evaluation pipelines in PyTorch** for systematic comparisons against strong baselines.
 - Analyzed results for **model architectures and loss functions** to improve segmentation performance.
- *Under Dr. Muhammad Tahir (Jun 2023-Jun 2024):*
 - Refined an **unrolled optimization algorithm for matrix completion** to improve baseline model ([convmc-net](#)).
 - Developed a novel extension to handle settings involving **Gaussian mixture model noise** ([ConvHuberMC-Net](#)).
 - Reproduced and evaluated prior methods, conducting **state-of-the-art benchmarking comparisons**.
 - Documented methodology and results in a **comprehensive technical report** ([report](#)).

PUBLICATIONS

- Kristo Nugraha Lian, **Nehal Ahmed Shaikh**, Di Hou, Xingyu Xie, Kim-Chuan Toh. *McADMM: A Multi-Clique Augmented Lagrangian-Based Algorithm for Large-Scale Sparse SDPs with Bound Constraints*. arXiv:2610.01503. Preprint.
- Talha Ahmed, **Nehal Ahmed Shaikh**, Hassan Mohy Ud Din. *Wave-GMS: Lightweight Multi-Scale Generative Model for Medical Image Segmentation*. ICASSP 2026.

PROJECTS

- **Executable Euclid:** Developed an exact-arithmetic computational geometry system that executes and verifies all propositions from Euclid's *Elements*, generates construction diagrams and dependency analyses, and publishes the results as an interactive site (code, site).
- **Three Stacks in Series:** Built a SAT-based combinatorial search and independent proof-checking pipeline that found a length-22 unsortable permutation, improving the previously published upper bound of 38; produced replayable witnesses and machine-checkable DRAT certificates (code).
- **Blindspot:** Built a reproducible Python pipeline that ingests and validates UN and World Bank data, measures cross-country data-coverage gaps, performs temporal and robustness analyses, and generates an interactive static atlas (code, site).
- **Speech Recognition and Translation System For Medical Communication:** Combined speech recognition, machine translation, and TTS functionality, selecting relevant vector databases, fine-tuning large language models (LLMs), and employing RAG pipelines to enable users to receive real-time state-of-the-art medical advice and information (see: [repository](#)).
- **Air Pollution & Academic Performance:** Explored the relationship between air pollution and academic performance in Pakistan to know whether there exists any relationship between the two and to predict academic performance scores based on a customized synthesis of ASER (education) and NASA (climate) data (see: [blog](#)).

EDUCATION COURSEWORK

BSc Economics-Mathematics (Computer Science minor) from Lahore University of Management Sciences. Reinforcement Learning, Generative AI, Deep Learning, Machine Learning, Signal Processing, Convex Optimization, Numerical Analysis, Differential Equations, Real Analysis, Statistics, Econometrics, Game Theory.

HONORS

Placed four times on Dean's Honor List for excellent academic performance and ultimately graduated with distinction (3.72/4 CGPA).

SKILLS

Programming languages (Python, C/C++, R, SQL, Julia), machine learning (PyTorch, TensorFlow, LightGBM, XGBoost, Stan, ONNX), LLMs (Google ADK, DeepEval, llama.cpp, GGUF, TFLite, prompt engineering, LLM evaluation), tools (Docker, Git/GitHub, Linux, Superset, LaTeX).